Damaged Building Detection From Post-Earthquake Remote Sensing Imagery Considering Heterogeneity Characteristics

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***Abstract*— Damaged building detection from remote sensing imagery helps to quickly and rapidly assess losses after an earthquake. In recent years, deep learning technology has become a favorable tool for remote sensing image information detection. Based on the characteristics of damaged buildings in remote sensing images, in this article, a framework for damaged building detection that considers heterogeneity characteristics is proposed. First, a local–global context attention module is proposed to improve the feature detection ability of the network, which can extract the features of damaged buildings from different directions and effectively aggregate global and local features. In addition, the module takes the correlation between feature maps at different scales into account while extracting information. Second, a feature fusion module with self-attention is established to replace the simple connection between the encoding and decoding processes, which improves the detail feature recovery ability of the network during the upsampling process. Finally, to fully aggregate semantic and detailed features at different scales, a multibranch auxiliary classifier is established by adding two separate branches in the prediction stage. The effectiveness of the proposed approach is verified based on the data from the 2010 Haiti earthquake, and comparisons with three object-oriented methods and 16 existing excellent deep learning models are performed. The intersection over union (IOU) increase of 0.03%–7.39% is achieved using the proposed approach compared with excellent deep learning models.**

***Index Terms*— Convolutional neural network (CNN), damaged building detection, heterogeneity characteristics, remote sensing imagery.**

I. Introduction

EARTHQUAKES can cause substantial damage in a short time. It is of great significance to carry out emergency rescue by capturing the distribution of a disaster in real time after an earthquake [1], [2], [3]. In recent years, remote sensing images have gradually become the main data for disaster detection in the process of emergency rescue due to their advantages of wide coverage, large amount of information, and short processing times. Using remote sensing images to interpret and evaluate damaged buildings can effectively improve the efficiency of disaster management [4], [5]. Therefore, it is of great significance to quickly and accurately extract information from damaged buildings after an earthquake.

The existing methods for damaged building detection using optical remote sensing images include the change detection method for multitemporal remote sensing images before and after an earthquake and damaged building detection directly from post-earthquake images [6], [7]. However, multitemporal evaluation methods can be difficult to apply in practice because of the influence of data acquisition, sensor revisit cycles, shooting angle, and time [8], [9]. Therefore, in existing studies, damaged building features have been directly extracted from post-earthquake images. The methods of damaged building detection using post-earthquake images include visual interpretation [10], pixel calculation [11], object-oriented [12], [13], and machine learning [14]. The visual interpretation method classifies damaged buildings in remote sensing images by visual observation and interpretation. Although this method can prevent the risk faced by manual field investigation and has high accuracy, it requires considerable manpower and material resources, and the knowledge learned from discrimination cannot be used repeatedly. To improve the efficiency, the pixel calculation method has been applied to extract features from damaged buildings. In this method, spectral, textural, and morphological features are extracted for modeling to obtain damaged building information from remote sensing images. For example, Aoki *et al.* [15] defined the characteristics of damaged areas according to hue, saturation, brightness, and edge strength and carried out the automatic detection of seismic damage on buildings using high definition television (HDTV) images from Kobe, Japan, in 1995. Li *et al.* [16] analyzed the Kashi earthquake in Xinjiang, China, by combining texture features and morphological features and extracted the damaged building area. However, the pixel calculation cannot make full use of damaged building information, including texture, morphology, and context, which restricts the accuracy of damaged building detection. In addition, the surface situation of post-earthquake images is complex, and there are many

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mixed pixels. The pixel calculation method lacks the ability to extract the relationship between pixels, resulting in serious "salt and pepper" phenomena in the detection results.

The object-oriented method was then proposed to solve this problem. For example, Wang *et al.* [17] used the object-oriented method to classify damage from the remote sensing aerial images of the Wenchuan earthquake. These researchers analyzed image texture and shape and their relationship information from remotely sensed images, and the results showed that the classification accuracy was significantly higher than that of traditional pixel classification methods [17]. Li *et al.* [18] established a feature set by analyzing image features, such as spectral information, texture, shape, and context, to identify buildings damaged in the 2008 Wenchuan earthquake by using the object-oriented method. Zhao *et al.* [13] proposed a multifeature combination segmentation method for damaged building detection by introducing texture features into the spectral and morphological features of remote sensing images. Although the object-oriented method alleviates the shortcomings of traditional pixel classification methods, the segmentation scale of the image needs to be determined by continuous exploration, and the features used for image classification still must be manually selected, i.e., the features cannot be automatically learned. Therefore, this method is unsuitable for large amounts of remote sensing images.

To automate the processing of damaged building detection, machine learning methods have been used to extract damaged building features from remote sensing images after an earthquake [19], [20], [21], [22], [23]. Machine learning discriminates the target by designing a feature detection function and uses the method of sample learning to extract the characteristics of objects in remote sensing images. For example, Taskin *et al.* [24] proposed a post-earthquake damaged area detection method based on the support vector selection. Janalipour and Mohammadzadeh [25] extracted the geometric features of damaged buildings, including the area, rectangular fitting, and convexity, and designed a decision-making system and a fuzzy inference system based on an adaptive network to determine the damaged buildings. However, machine learning is limited to shallow feature extraction, and its nonlinear expression ability is limited. Therefore, it cannot be used to extract the deep-level features of damaged buildings in remote sensing images.

With the development of deep learning technology [26], many neural networks, including recurrent neural networks (RNNs) [27], convolutional neural networks (CNNs) [28], and deep belief networks (DBNs) [29], have been proposed. Among them, CNNs have a good application effect in image classification and target recognition [30], [31]. Researchers have proposed many improved algorithms based on CNNs, such as fully convolutional network (FCN) [32], pyramid scene parsing network (PSPNet) [33], and DeepLabv3 [34], which have been applied to image semantic segmentation. At the same time, many neural networks have been used to extract information from remote sensing images [35], [36], [37], [38], [39], [40], [41].

On this basis, some studies have extracted damaged buildings using remote sensing images [42], [43], [44]. Ji *et al.* [45] proposed a CNN-based method for identifying damaged buildings in post-earthquake remote sensing images. Duarte *et al.* [46] used different resolution images combined with the CNN model to evaluate damaged buildings in images. However, this model can only be used to classify damaged building from images and cannot be used to perform detailed semantic segmentation. Jiang *et al.* [23] proposed a superpixel-based belief fusion model for damaged building detection. Bai *et al.* [47] used U-Net to evaluate the damage to buildings after the Tohoku earthquake and tsunami in 2011 and divided the damaged buildings into three categories: "washed away," "collapsed," and "survived". Ma *et al.* [48] used 0.5-m aerial images to detect damaged buildings after the Yushu earthquake by improving Inception-v3. Later, Ma *et al.* [49] detected damaged buildings after an earthquake based on an improved YOLOv3 model. Wheeler and Karimi [50] established a semantic segmentation model based on deep learning to evaluate the damage degree of damaged buildings after an earthquake.

Although these studies are notable, there are still great challenges in the damaged building detection process. First, the building characteristics of spectral information, texture, and morphology are destroyed after an earthquake, so feature detection is difficult. Neural networks obtain high-level abstract features from data through nonlinear expression, which makes it possible to extract the deep-level features of damaged buildings. Second, a damaged building may not completely collapse after an earthquake. As a result, the non-collapsed part of the building may resemble the undamaged building. Therefore, damaged building detection not only faces interference from the common background but also encounters interference from undamaged buildings. Finally, existing feature detection methods based on CNNs mostly originate from improvements in the existing semantic segmentation model from the computer vision field. However, unlike the ground objects in ordinary images, remote sensing images have more complex features. It is a challenge to design a semantic segmentation model suitable for damaged building detection according to the characteristics of damaged buildings in remote sensing images.

To overcome these problems, the analysis of the characteristics of damaged buildings in remote sensing images is performed in this article, and a network considering the heterogeneity characteristics of damaged buildings is proposed. Using the proposed method, the following four main contributions are made.

1) A deep learning model for damaged building detection is proposed by analyzing the heterogeneity of damaged building features in remote sensing images from four aspects: spectral information, texture, shape, and context. According to different characteristics of damaged buildings, different modules are established, including a local–global context attention module (LGAM) for feature detection, a feature fusion module with self-attention (FFS), and a multibranch auxiliary classifier. The results show that the proposed approach performs better than the existing competitive algorithms.

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2) Sufficient feature extraction is the key to obtaining high-accuracy results. An LGAM is established to fully obtain the heterogeneous characteristics of damaged buildings. The local features can be obtained while obtaining the global features on the same scale, and the global and local feature association can be enhanced by point multiplication. On this basis, a feature extraction module with a stronger correlation between layers is established in combination with ResNet101 [51] to improve the ability of the network to extract features at different scales during the feature extraction process.
3) We propose an FFS, which can enhance the correlation between encoding and decoding structures by self-learning and improve the recovery ability of the network for detailed features during the upsampling process.
4) We establish a multibranch auxiliary classifier, which adds two branch prediction structures based on multiscale aggregation. The classifier can enhance the understanding of the network at different levels and further improve the classification ability of the network in the final stage.

The following sections are organized as follows. The study area and heterogeneity characteristics of damaged buildings are introduced in Section II. The proposed approach is presented in detail in Section III. The experimental configuration and evaluation metrics are introduced in Section IV. The experimental results and discussion of this article are presented in Section V. Finally, the conclusion and future work are presented in Section VI.

II. Study Area and Heterogeneity Characteristic Analysis of Damaged Buildings

In this section, we first introduce the study area and dataset generation. Then, we describe the damaged buildings. Finally, the heterogeneity characteristics of damaged buildings in optical remote sensing images are analyzed in detail.

A. Overview of the Study Area

The area from an optical image taken after the Haiti earthquake on January 12, 2010, is selected to verify the proposed approach. The magnitude of the Haiti earthquake was 7.0, with a focal depth of 10 km. The death toll was 270 000, and 3.7 million people were affected. This is the strongest recorded earthquake in Haiti since 1770 and has brought considerable casualties and economic losses. The data are the open-source disaster images provided by DigitalGlobe (https://www.digitalglobe.com/ecosystem/open-data), and the satellite image is from the Maxar Satellite Group. The image resolution is 0.5 m, and the image consists of red, green, and blue bands. The time of image acquisition was the third day after the earthquake (January 15, 2010). The use of optical remote sensing images over a short period after an earthquake to extract the characteristics of damaged buildings can eliminate the interference of various factors. In addition, the overall arrangement of buildings in the image of the study area is regular, the separation zone between houses is narrow, and the houses are mainly low-rise buildings. The background objects in the remote sensing images include vegetation, bare land, mountains, and oceans. The coverage area mainly includes the Jacmel and Port-au-Prince areas. The label-making tool LabelMe is used to produce the dataset, which is annotated by manual visual interpretation, as shown in Fig. 1. Because the original image is too large to be directly input into the neural network for training, the complete data are reduced to $256 \times 256 \times 3$. Finally, a dataset that can be used for neural network training is formed. This dataset includes 2257 images for training, 270 images for verification, and 567 images for testing.

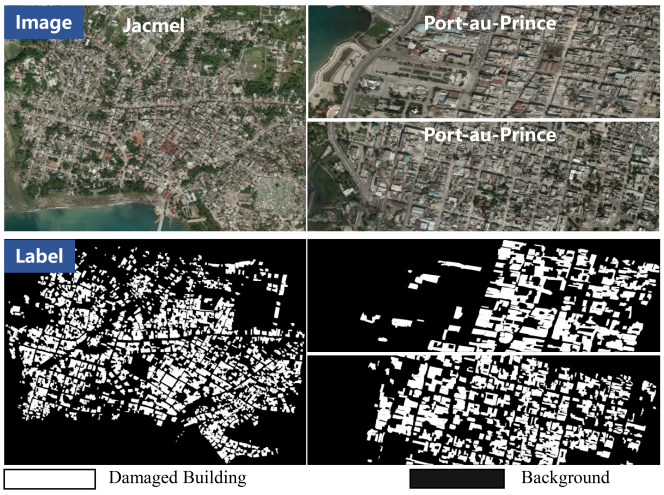

Fig. 1. Overview of the study area: optical remote sensing image and the corresponding ground-truth damaged building masks.

B. Description of the Damaged Buildings

Different earthquake intensities will cause different degrees of damage to buildings when an earthquake occurs. However, the remote sensing image is the response of the spectral characteristics of ground objects, and the recorded information can only reflect the surface conditions of damaged buildings; the internal structure of buildings cannot be reflected. Even in high-resolution images, the details of earthquake damage are difficult to detect. Therefore, according to the classification method of damaged buildings in the existing research [46], [49], in this article, the objects in the image are divided into two categories, including damaged buildings and background. Damaged buildings include the partial collapse and displacement of buildings [the red-framed area in Fig. 2(a)] and complete collapse of buildings [the red-framed area in Fig. 2(b)]; that is, buildings with obvious damage can be visually identified. Notably, the above categories are uniformly defined as damaged buildings without detailed classification. The background includes undamaged buildings and other objects in remote sensing images.

C. Heterogeneity Characteristic Analysis

The characteristics of damaged buildings show more complexity and diversity in terms of spectral information, texture,

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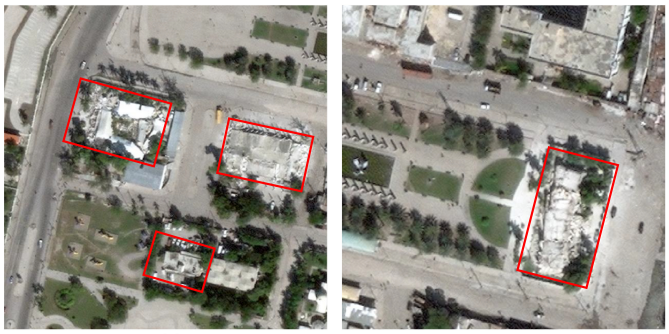

(a) (b)

Fig. 2. Description of damaged buildings: (a) partial collapse of buildings and (b) complete collapse of a building.

morphology, and context, which are called the heterogeneity characteristics of damaged buildings. The heterogeneity characteristics of damaged buildings are analyzed in detail and can mainly be divided into the following four aspects, as shown in Fig. 3. The first row is the pre-earthquake image, and the second row is the post-earthquake image.

1) *Large Spectrum Difference*: The roofs of buildings often have a similar material and color before an earthquake. Different from the spectral characteristics of intact buildings, the mutual doping between the collapsed walls and gravels increases the spectrum value noise of the buildings after an earthquake. In addition, with increasing building damage, the spectrum complexity of the image will also increase, as shown in Fig. 3(a).
2) *Uneven Texture Features*: The texture structure of the roof of a building is regular before an earthquake, and it is easy to distinguish the building. However, after an earthquake, the texture is chaotic along different directions, and the surface features are uneven and difficult to distinguish, as shown in Fig. 3(b).
3) Morphological characteristics are destroyed. The geometric shapes of buildings before an earthquake are regular, e.g., rectangular-shaped, square-shaped, I-shaped, L-shaped, or a combination of multiple shapes. In contrast, the morphological characteristics of buildings are destroyed after an earthquake, and the original regular spatial distribution is also disrupted, as shown in Fig. 3(c).
4) Context features are more complex. The difference in context features is related to the impact of the collapse on the surrounding environment. For example, roads or vegetation are separated between buildings in residential areas before an earthquake, and the boundaries between buildings and surrounding objects are identifiable. The broken rubble after an earthquake can block streets and vegetation, making buildings and surrounding ground objects difficult to distinguish from each other, as shown in Fig. 3(d).

III. Methods of Damaged Building Detection in Remote Sensing Imagery

Based on the heterogeneity characteristics analysis of a damaged building, a network considering heterogeneity characteristics to extract features from damaged buildings is proposed, as shown in Fig. 4. An LGAM for feature detection is established, which obtains the local and global information in the encoding stage. This module is inspired by the research ideas in [51], [52], and [53], as shown in Fig. 4(a). In addition, the module can enhance the correlation between different scales.

The decoding stage includes an FFS and a multibranch auxiliary classifier, as shown in Fig. 4(b). First, to fully fuse the features extracted from various levels, an FFS is established; this module can transfer the rich information extracted during the encoding stage to the decoding stage and improve the detailed feature recovery ability of the network in the decoding process. This module is inspired by the idea of attention [54]. Second, we establish a multibranch auxiliary classifier. This classifier is added to two auxiliary branches based on multiscale output, which can further strengthen the comprehensive mapping of the network to different levels of features. After each round of training, the parameter of the model is continuously updated through backpropagation until the optimal solution is formed, as shown in Fig. 4(c).

A. LGAM for Feature Detection

We establish an LGAM, which consists of three parts, to extract the complex heterogeneous characteristics of the damaged buildings. First, a local detail awareness module is established to fully obtain the complex local features of the spectrum, texture, and context of damaged buildings, as shown in Fig. 5(a). Second, the global context information is further extracted through the attention structure, and it is weighted with local features by the local detail awareness module to improve the feature detection ability of the network, as shown in Fig. 5(b). Finally, a feature detection module with a stronger correlation between adjacent layers is formed by combining it with ResNet101 [51], as shown in Fig. 5(c).

1) *The Local Detail Awareness Module* [*Fig. 5(a)*]: Through the above analysis, the damaged building area has more complex characteristics. Therefore, comprehensive detail perception is realized through convolution operations in different directions, which is inspired by Pan *et al.* [52] and Hu *et al.* [53]. We aggregate detailed information in four directions and generate maps of weights to combine the features for different directions, as shown in the following equation:

$$W_{\mathrm{ULDR}} = \mathrm{Con}(W_{\mathrm{U}}, W_{\mathrm{L}}, W_{\mathrm{D}}, W_{\mathrm{R}}) \tag{1}$$

where W_{ULDR} is the result of the local detail awareness module. $W_{\mathrm{U}}, W_{\mathrm{L}}, W_{\mathrm{D}}$, and W_{R} represent the weight information obtained in different directions. Con denotes the concatenation.

1) *Global Context Information Detection* [*Fig. 5(b)*]: We further obtain the global context information at the same level based on batch normalization and the sigmoid activation function. Then, the weighted operation is carried out by multiplication to strengthen the correlation between the detailed information and the global information, which helps to obtain more comprehensive

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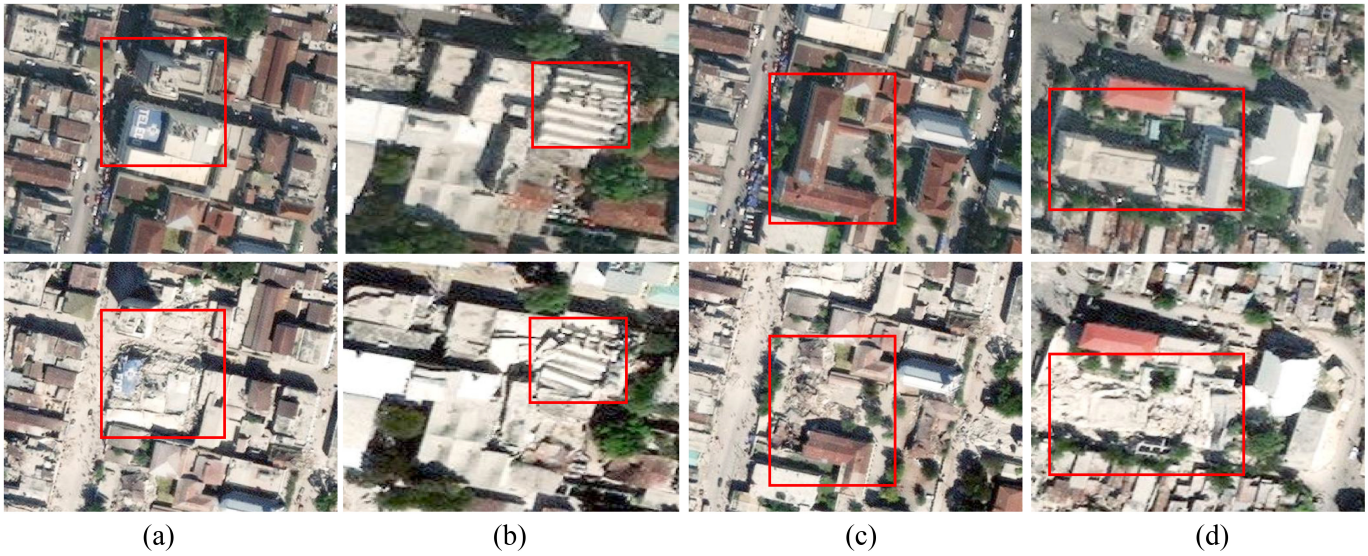

Fig. 3. Heterogeneity characteristic analysis of damaged buildings in remote sensing images: (a) large spectrum difference; (b) uneven texture features; (c) morphological characteristics are destroyed; and (d) context features are more complex.

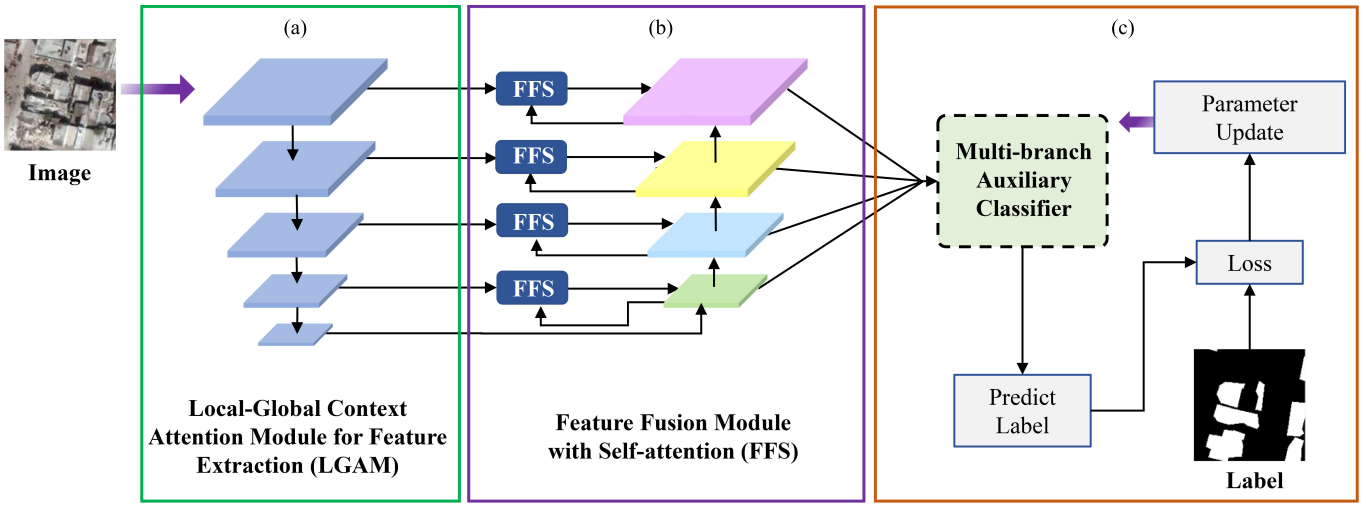

Fig. 4. Framework of our proposed network: (a) encoder; (b) decoder; and (c) classifier.

features of the damaged buildings at the same scale. The operation can be formulated as shown in the following equation:

$$\text{Output}_{\text{AB}} = \text{Conv}_{1*1}(W_{\text{ULDR}}) * \text{Sigmoid}\big(\text{Conv}_{(1*1)}(T)\big) \quad (2)$$

where $\text{Output}_{\text{AB}}$ is the output feature map. Conv_{1*1} represents the convolution operation of 1×1. T represents the input feature map. Sigmoid represents the sigmoid activation function. W_{ULDR} is the result of the local detail awareness module.

1) *Interlayer Feature Association Module* [*Fig. 5(c)*]: In the process of continuous downsampling, the semantic feature is continuously enhanced while the detailed feature is constantly lost. There is often only one pooling structure between different levels in the existing network, which causes the feature association between layers to be weak. To enhance the feature association between layers, we establish an interlayer feature association module. First, the pooling structure is replaced by a convolution structure with a stride of 2 to avoid feature loss. Second, the ResNet101 structure is used to obtain the deep information of damaged buildings as a separate branch. Finally, the two branches form a feature extraction structure with closer interlayer features. Additionally, the fusion result is used as the input feature map of the next layer, which can enhance the feature connection of different scales in the encoding stage and improve the feature extraction ability in the downsampling process. In addition, the dropout function is added after each module of the LGAM to avoid overfitting.

B. Feature Fusion Module With Self-Attention

The semantic features at different levels are obtained by continuous downsampling in the encoding stage of the network. At the same time, the feature details are lost by

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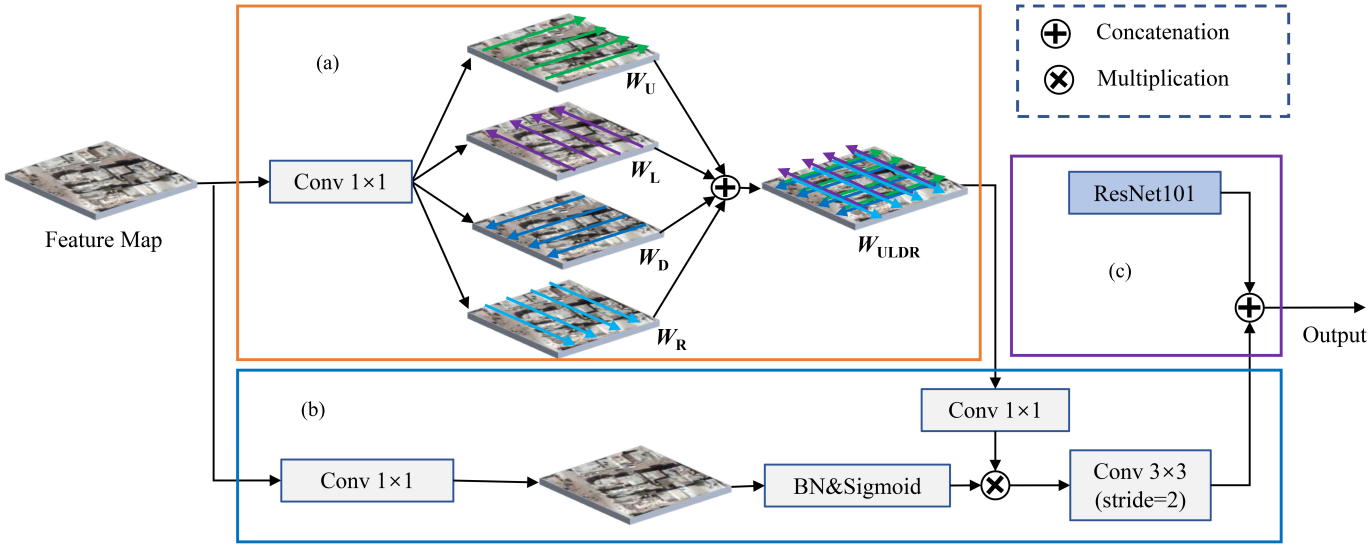

Fig. 5. Structure of an LGAM for feature extraction: (a) local detail awareness module; (b) global context information extraction; and (c) interlayer feature association module.

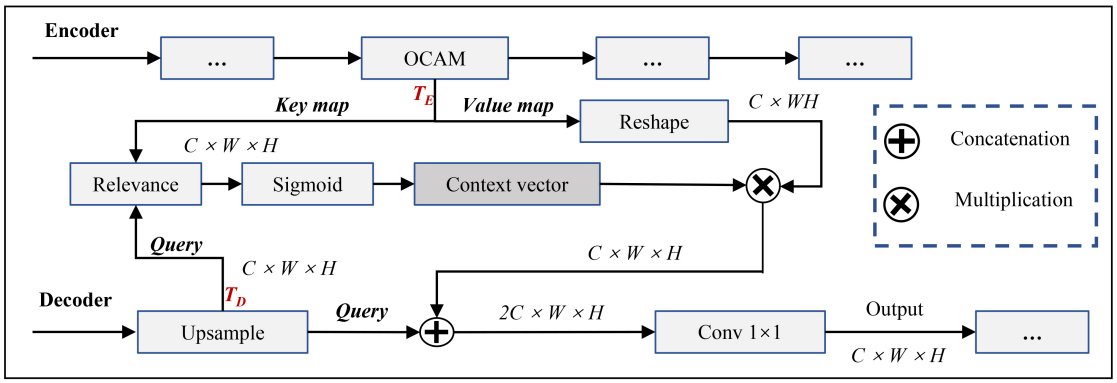

Fig. 6. Structure of a feature fusion with the self-attention module.

continuous downsampling in the encoding stage. The upsampling process in the decoding stage can be used to recover these features. However, the detailed features cannot be sufficiently recovered through only the upsampling operation. To solve this problem, many researchers have fused the upsampling feature map with the same scale feature map in the encoding stage to recover more detailed features [55]. However, in the existing methods, only a simple channel splicing operation is performed on the encoding and decoding layers at the same scale. As a result, there is no self-learning ability, and the feature fusion ability is limited. Therefore, to improve the learning ability in the fusion process, we propose an FFS, as shown in Fig. 6. The encoding and decoding output feature maps are represented by T_{E} and T_{D}, respectively. Additionally, the feature tensor Key (K), Query (Q), and Value (V) required for self-attention are defined as $K = V = T_{\mathrm{D}}$ and $V = T_{\mathrm{E}}$. The method is mainly divided into three steps.

1) The relevance of feature maps between the encoding and decoding stages at the same scale is calculated by the point multiplication of the Key and Query. Then, the sigmoid function is used for normalization to obtain the optimized context relation vector, as shown in the following equation:

$$\text{Context vector} = \text{Sigmoid}\ (\text{Key} * \text{Query}) \tag{3}$$

where Contextvector is the optimized context relation vector, Key represents T_{E}, and Query represents T_{D}.

1) The obtained context relation vector is further correlated with the encoding output result Value after reshaping according to the typical attention idea. The weight mapping optimization of the output of the encoding stage is carried out again. This optimization helps the network capture the key features of detailed information by self-learning and reduces noise interference. Then, the feature weight mapping after the second optimization is connected with the output feature map in the decoding stage. Compared with the traditional simple connection, the connection method based on self-attention can be implemented to improve the attention of the network to the key areas. Then, a 1×1 convolution is used to maintain the number of dimensions in the continuous upsampling process and reduce characteristic redundancy.

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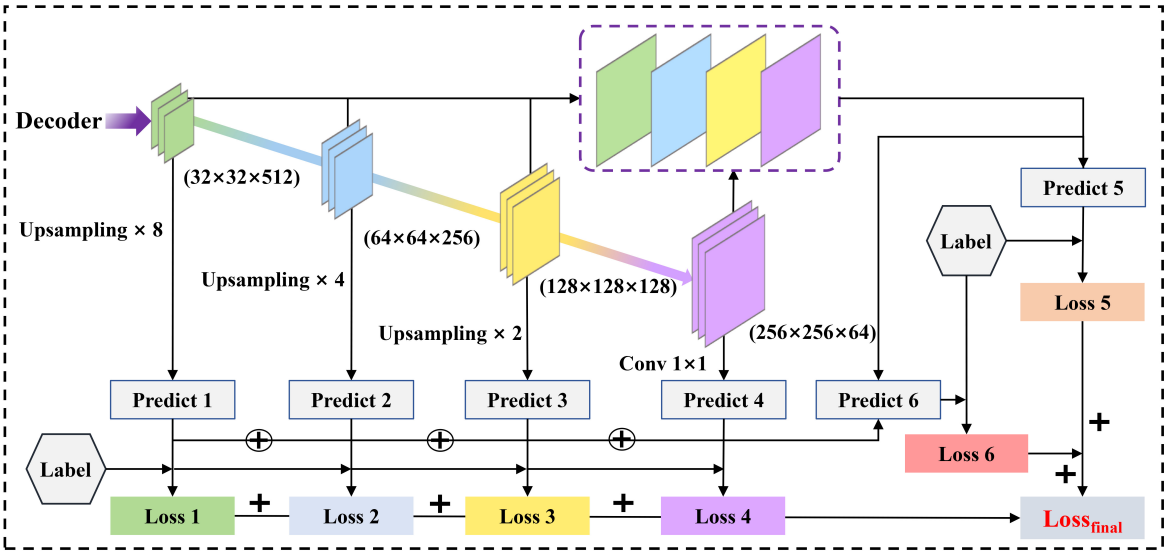

Fig. 7. Structure of the multibranch auxiliary classifier.

This operation can be formulated as shown in the following equation:

$$\text{Output} = \text{Conv}_{(1*1)} \times (\text{Con}(\text{Query}, \text{Mul}(\text{Context vector} * \text{Reshape}(\text{Value})))) \quad (4)$$

where Output is the result of the feature fusion module, which is used for the next upsampling. $\text{Conv}_{(1*1)}$ represents the 1×1 convolution operation, Con denotes the concatenation, Mul denotes the multiplication, Query denotes T_{D}, and Value denotes T_{E}.

C. Multibranch Auxiliary Classifier

In the process of semantic segmentation, the low-level feature map has high resolution and contains more detailed information and less semantic information. The high-level feature map has low resolution and contains less detailed information and more semantic information. The final classifier is a key step to aggregate low-level features and high-level features. A multibranch auxiliary classifier is established to obtain detailed and semantic information of damaged buildings. Based on the existing multiscale prediction structure, two branches are separately added to further strengthen the expression and understanding ability of the network at the final classification stage for different scales, as shown in Fig. 7.

First, our classifier includes the prediction of all scales in the upsampling process, including the outputs of the 1, 1/2, 1/4, and 1/8 scale layers (Predict 1, Predict 2, Predict 3, and Predict 4, respectively). Second, although the details of the upsampling process will continue to recover, the loss by downsampling is inevitable. We superimpose the prediction results of the four scales and aggregate all the maps (Predict 5) to obtain denser details. Finally, the final feature integration is conducted on the results of the above five outputs (Predict 6), in which all detailed information and semantic information are aggregated. In addition, two auxiliary branches are added as the feature output based on the original network to avoid the increase in network parameters and ensure the improvement in accuracy. The role of each branch in the classifier is carefully analyzed in Section V-F.

IV. Experimental Configuration and Evaluation Metrics

A. Experimental Configuration

1) Training Environment Description: All training and testing platforms for the experiment are implemented on the Windows 10 system. The network model is built based on the PyTorch deep learning framework. The experiment is performed with a single Nvidia GeForce RTX 3090 24G GPU and an AMD Ryzen 7 5800X 8-Core Processor 3.80-GHz processor (CPU).

2) Hyperparameter Settings: The internal parameters of the neural network are iteratively obtained by training the model, and some hyperparameters need to be artificially set before training. In our experiment, the parameters in the training process are set as follows: the Batch_size is set to 12, the network optimizer is AdaMax, and the initial learning rate is set to 0.0005. The parameters trained by ResNet101 in ImageNet are used as the initial weight to improve the stability and generalization of the model. In the process of training, we monitor the loss value of the validation data and save the model only when the loss value of the validation data of this round is lower than that of the previous round. In addition, data augmentation is used in the process of training to avoid overfitting.

The loss function is used to evaluate the difference between the probability distribution obtained by the current training and the real distribution. In this article, we establish a loss function on the basis of the binary cross entropy (BCE) [see (5)] [56] according to the setting of the multibranch auxiliary classifier, as shown in (6). The loss curve in the training process is shown in Fig. 8. It can be seen from Fig. 8 that the model loss tends

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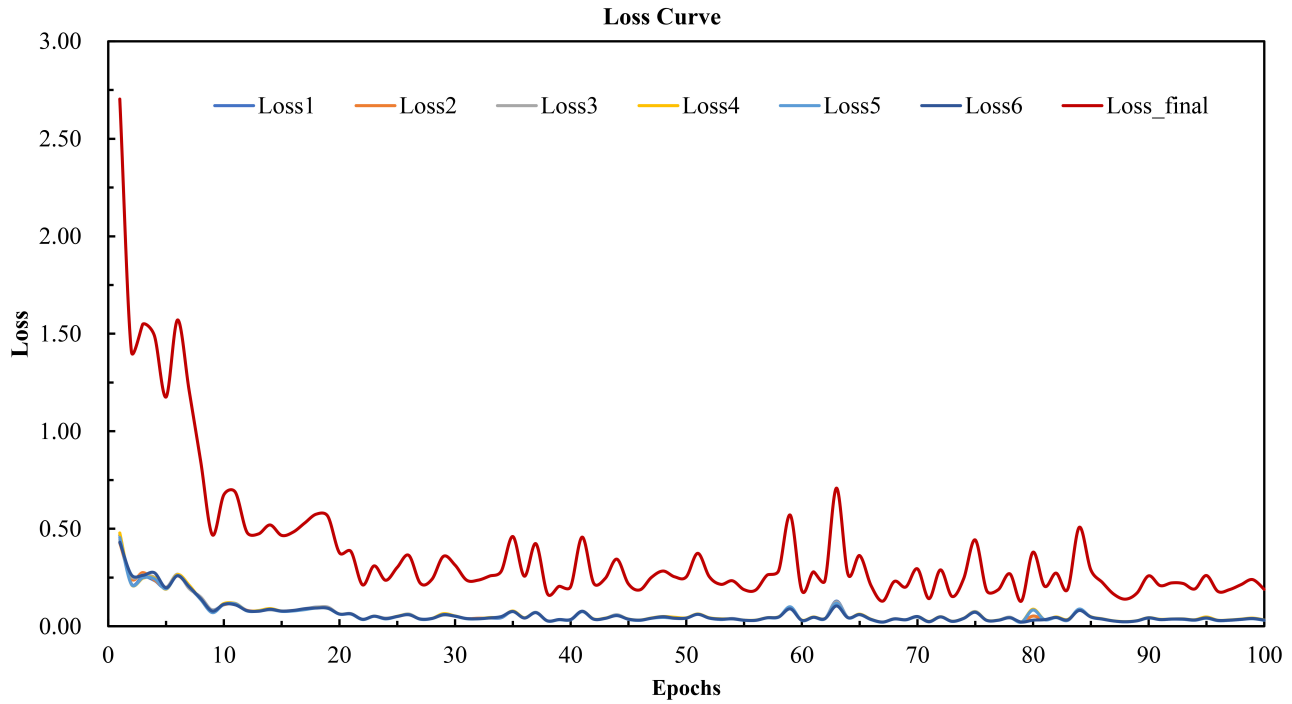

Fig. 8. Loss curve of our method in the training process.

to stabilize after approximately 100 epochs

$$\text{Loss} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log p_i + (1-y_i)\log(1-p_i)\right] \quad (5)$$

$$\text{Loss_final} = L_{\text{loss1}} + L_{\text{loss2}} + L_{\text{loss3}} + L_{\text{loss4}} + L_{\text{loss5}} + L_{\text{loss6}} \quad (6)$$

where Loss represents the BCE loss function, which is the difference between the result probability distribution predicted by the model and the real value probability distribution. $y_i = \{0,1\}$ represents the pixel category of the label. $p_i \in (0, 1)$ represents the probability of damaged buildings. N is the total quantity of input pixels. L_{loss1} to L_{loss6} represent the loss value of each branch in the multibranch auxiliary classifier. Loss_final represents the final loss value in the process of training.

B. Evaluation Metrics

To quantitatively evaluate the proposed approach and compare it with the existing methods, we select five commonly used semantic segmentation criteria to evaluate our method, including intersection over union (IOU) [see (7)], overall accuracy (OA) [see (8)], F1-score [see (9)], precision (P) [see (10)], and recall (R) [see (10)]

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \quad (7)$$

$$\text{Overall accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FN}+\text{FP}+\text{TN}} \quad (8)$$

$$\text{F}_1 = 2 \times \frac{P \times R}{P + R} \quad (9)$$

$$P = \frac{\text{TP}}{\text{TP}+\text{FP}} \text{ and } R = \frac{\text{TP}}{\text{TP}+\text{FN}} \quad (10)$$

where TP is the true positive, which represents the number of correctly extracted damaged building pixels, FP is the false positive, which represents the quantity of erroneously damaged building pixels, TN is the true negative, which represents the total quantity of correctly extracted undamaged building pixels, and FN is the false negative, which is the quantity of missed damaged building pixels.

V. Experimental Results and Discussion

In this section, the experimental results of our method are described and compared with the state-of-the-art (SOTA) semantic segmentation networks in detail. Then, we discuss the proposed approach in detail, including the stability, feasibility, anti-interference, transferability, and ablation experiments. Finally, we analyze the advantages and limitations of the proposed approach in detail.

A. Damaged Building Detection Result

To comprehensively verify the effectiveness of the proposed method, three object-oriented methods and 16 excellent semantic segmentation methods were used to perform accuracy comparison experiments with the proposed approach under the same conditions. These methods included not only the classic semantic segmentation methods widely used in recent years but also some SOTA deep learning methods. We used the pretraining weight for transfer learning to ensure that the comparison methods obtained the best result.

1) Quantitative Analysis: During the experiment using object-oriented methods, we referred to the optimal scale selection method in [57]. After repeated debugging, the segmentation scale was finally set to 50, the edge-based segmentation algorithm was implemented, the merge scale was set to 80, the full lambda-schedule algorithm was utilized,

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TABLE I
DAMAGED BUILDING DETECTION RESULT COMPARISON WITH OBJECT-ORIENTED METHODS AND SOTA SEMANTIC SEGMENTATION NETWORKS

	Methods	IOU (%)	P (%)	R (%)	F1	OA (%)
Object oriented methods [57]	KNN	46.13	52.40	79.42	63.14	66.04
	SVM	46.79	53.53	78.80	63.75	67.18
	PCA	47.30	49.77	87.27	63.39	63.08
Deep learning methods	FCN8S [32]	61.48	68.12	86.31	76.14	83.65
	FCN32S [32]	62.15	70.86	83.48	76.65	83.46
	FCN16S [32]	62.69	71.61	83.42	77.07	83.67
	ICNet [58]	63.85	71.53	83.62	77.10	84.49
	CGNet [59]	64.06	72.74	84.29	78.09	84.36
	ESPNetv2 [60]	65.44	76.69	81.68	79.11	84.48
	DeepLabv3 [34]	65.95	74.16	84.63	79.05	85.33
	DUNet [61]	66.44	76.58	83.38	79.84	85.41
	Denseaspp [62]	66.56	77.42	82.58	79.92	85.09
	PSPNet [33]	66.82	79.69	81.58	80.62	85.03
	Deeplabv3_plus [63]	66.99	78.82	82.80	80.76	85.31
	ENCNet [64]	67.40	77.83	83.41	80.52	85.57
	DANet [65]	68.13	80.22	81.75	80.98	85.71
	OCNet [66]	68.28	79.22	82.85	80.99	85.77
	FDRNet [67]	68.25	79.93	81.34	80.63	85.75
	SIPE [68]	68.84	79.66	82.62	**81.11**	**85.97**
	Ours	**68.87**	79.78	82.31	**81.03**	**85.83**

and the texture kernel size was set to 5. Then, an experiment was performed with three machine learning models provided in ENVI, including the K-nearest neighbor (KNN), support vector machine (SVM), and principal component analysis (PCA) models, as shown in Table I. The results show that the accuracy of the object-oriented methods was low, and the highest IOU that could be achieved in the same research area was only 47.30%. In addition, precision and recall were extremely unbalanced. A high recall indicates that there were many false positives in the results, that is, a large number of undamaged buildings were detected as damaged buildings. This is because the machine learning method can only extract linear features, and the expression ability of nonlinear features is limited. Therefore, these methods are unable to extract deep semantic features from images. In addition, during the experiment, the segmentation scale needed to be continuously debugged and manually verified, which does not meet the emergency rescue that needs after an earthquake.

Then, we quantitatively analyzed the performance of 16 methods on the testing dataset, as shown in Table I. Compared with other methods, a 0.03%–7.39% increase in IOU was achieved using the proposed method. An improvement in IOU is extremely difficult as the IOU is the most stringent metric in the semantic segmentation task. These results occurred because our method is an adaptive improvement based on the heterogeneity characteristics of damaged buildings in remote sensing images. However, the proposed method did not have an advantage in all metrics; F1 and OA can maintain an accuracy similar to that of Self-supervised Image-Specific Prototype Exploration (SIPE). In general, the proposed approach is competitive, even compared with the latest algorithm.

2) Qualitative Analysis: We compared the accuracy of all methods in Section V-A1. To show the performance of the method more intuitively, we chose seven methods with better accuracy for further analysis. These methods are also excellent neural networks in the field of semantic segmentation. First, we compared the IOU and OA metrics, as shown in Fig. 9.

On this basis, we selected the testing images of different damage degrees and areas for visual analysis, as shown in Fig. 10. The density of damaged buildings increases from the first row to the fifth row of images in Fig. 10. It can be seen from the first- and second-row images in Fig. 10 that for the damaged areas of buildings with small density and obvious characteristics, all methods detected the main areas of damaged buildings. However, our method yielded a low number of false negative and false positive results. In addition, the boundary detection result of our method was clearer. In particular, Deeplabv3_plus showed high false positive rates, especially for images in the second row. With the increasing density of damaged buildings, the advantages of our method become gradually more obvious, as shown in the third–fifth rows in Fig. 10. Although all methods were affected by varying degrees, especially with respect to the number of false positive results, and yielded reduced accuracy, our method still detected the main areas of damaged buildings and had a fewer number of false negative and false positive results. The results of qualitative analysis show that the heterogeneity characteristics are crucial and effective when extracting features from images of damaged buildings.

B. Stability Analysis

It is an important challenge to find a balance between precision and recall in the task of semantic segmentation. Therefore, we further analyzed the difference between the precision and recall of the proposed method and the comparison models, as shown in Fig. 11. Two different turning

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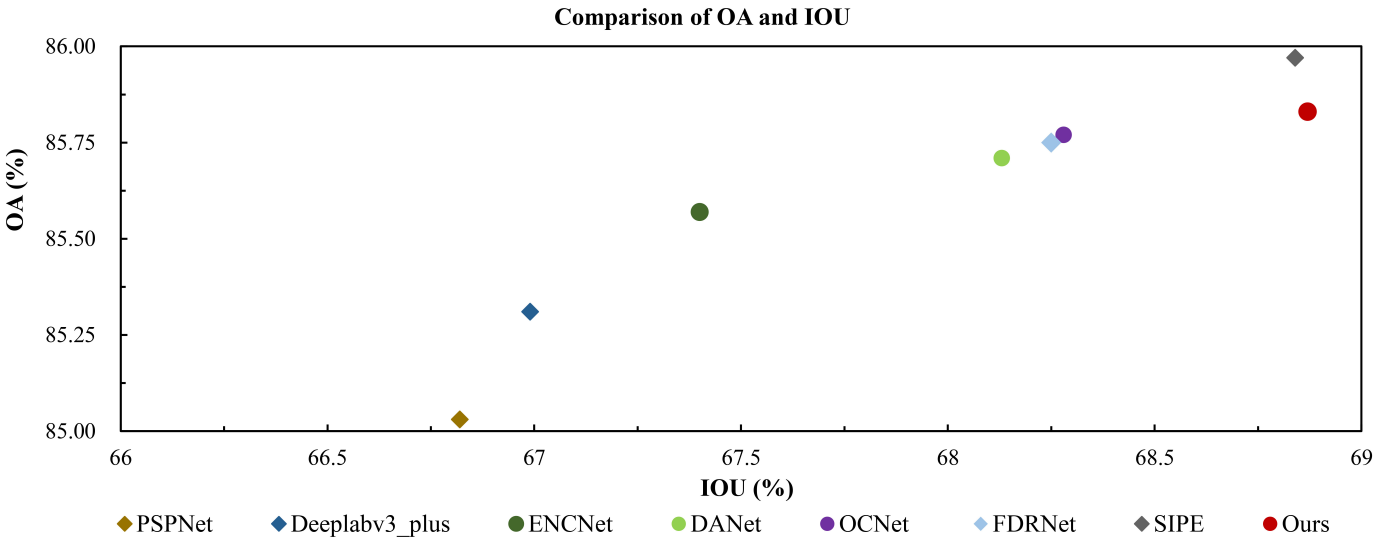

Fig. 9. Comparison of IOU and OA for the SOTA methods.

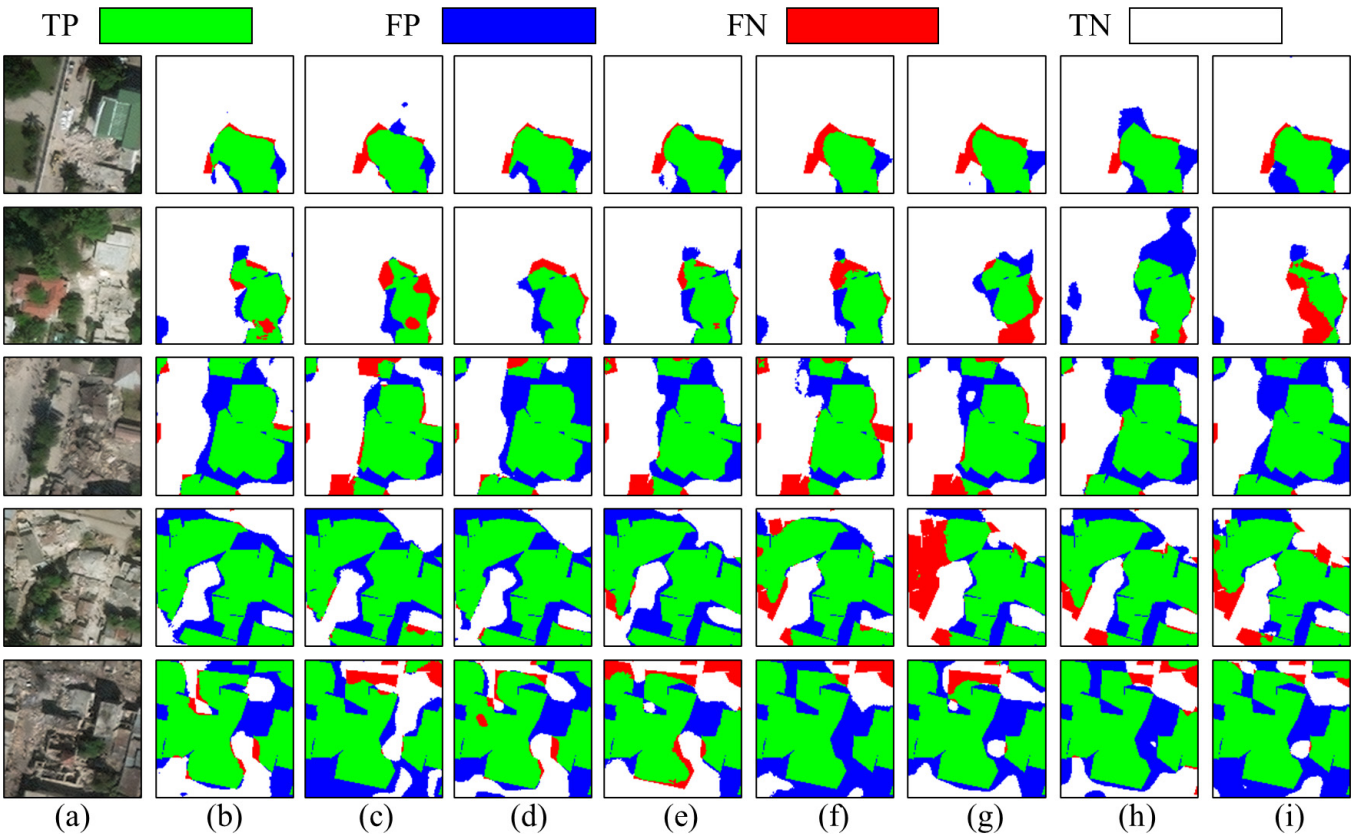

Fig. 10. Visual comparison of our method versus other SOTA damaged building detection methods: (a) original image; (b) our method; (c) FDRNet; (d) SIPE; (e) OCNet; (f) DANet; (g) ENCNet; (h) Deeplabv3_plus; and (i) PSPNet.

curves were used to represent precision and recall, and the box between the curves represents their difference. It can be seen from Fig. 11 that the difference between the precision and recall of the proposed approach is 2.53%, and the balance ability of our method is better than that of most other methods. In contrast, the differences between the precision and recall values of the PSPNet, dual attention network (DANet), and feature decomposition and reweighting network (FDRNet) methods are less than our method, with values of 1.89%, 1.53%, and 1.41%, respectively. On this basis, we further analyzed other evaluation metrics from Table I. Improvements in IOU, F1, and OA of 0.62%–2.05%, 0.05–0.41, and 0.08%–0.80%, respectively, were achieved using our method.

To further verify the stability of the proposed approach, we conducted ten independent Monte Carlo experiments on

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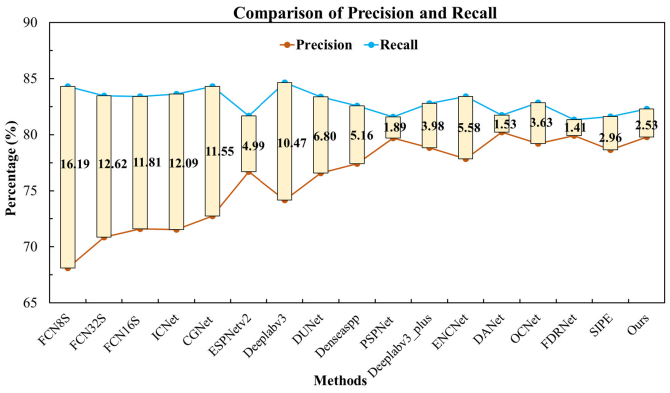

Fig. 11. Comparison of the differences between precision and recall.

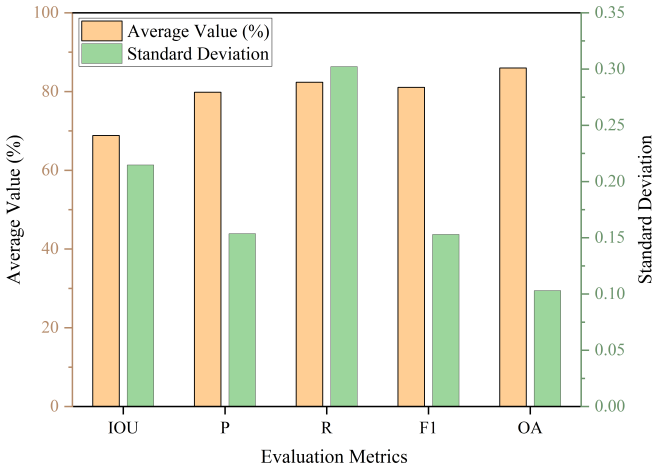

Fig. 12. Average value and standard deviation of Monte Carlo operations.

our method. The Monte Carlo experiment is a random sampling technique that can test the model in a random way to judge whether the model is stable. The experimental method involves adding dropout to the LGAM, setting the probability to 0.5, and randomly discarding the obtained features. The accuracy results of IOU, F1, and OA are calculated, and the average accuracy and the standard deviation of each index are calculated, as shown in Fig. 12. Fig. 12 shows that our method can still achieve high accuracy in the independent Monte Carlo experiment. In particular, each metric has a small standard deviation, in which IOU is 0.21, F1 is 0.15, and OA is 0.10, which further proves the stability of this method.

C. Feasibility Analysis

Considering the urgency of emergency rescue, in addition to accuracy, the feasibility of the method is also an important standard to evaluate its performance. Therefore, we calculated statistics on the training time and testing frame rate of all methods, as shown in Table II. It can be seen in Table II that the training time of the FCN was the shortest, and its detection speed was the highest. However, the FCN model achieved the lowest accuracy due to its simple model structure. The training time of our method was approximately 2.52 h, and its speed

TABLE II
TRAINING TIME AND FPS COMPARISON OF DIFFERENT METHODS

Methods	Training Time (h)	PFS
FCN8S [32]	**0.53**	**75**
FCN32S [32]	**0.53**	**74**
FCN16S [32]	**0.54**	**79**
ICNet [58]	2.03	23
CGNet [59]	1.05	65
ESPNetv2 [60]	0.94	68
DeepLabv3 [34]	2.62	30
DUNet [61]	2.48	32
Denseaspp [62]	2.32	26
PSPNet [33]	2.67	31
Deeplabv3_plus [63]	2.91	32
ENCNet [64]	2.73	30
DANet [65]	2.77	29
OCNet [66]	2.59	29
FDRNet [67]	1.96	27
SIPE [68]	2.34	25
Ours	2.52	30

TABLE III
COMPARISON OF TEST SPEED UNDER DIFFERENT CALCULATION CONFIGURATIONS

Calculation configuration	FPS
Nvidia GeForce RTX 3090 with 24G of onboard memory	30
Nvidia GeForce RTX 2070 with 8G of onboard memory	23
Nvidia GeForce RTX 3060 with 6G of onboard memory	18
Nvidia GeForce GTX 1060 with 6G of onboard memory	10

was approximately 30 frames per second (FPS). The training and testing times of our method are on the same order of magnitude as those of the existing models, but the highest accuracy was achieved using our method, which further proves its effectiveness.

To further verify the efficiency of our method, we tested our method on four different calculation configurations, and the results are shown in Table III. The results indicate that our method could be applied to different devices, and the detection speed on different devices of our method was 30, 23, 18, and 10 FPS. Notably, the detection speed reached 10 FPS even in an environment with poor calculation conditions. This further verified that the method in this article exhibits good feasibility.

D. Anti-Interference Analysis

Different from common object detection in remote sensing images, damaged building detection is a more challenging task. First, the detection of damaged buildings will be disturbed by similar features in the background of remote sensing images. In addition, the damaged building is generated on the basis of the original building. Although differential changes will occur after damage, the damaged building detection will still be disturbed by undamaged buildings. Therefore, we further analyzed the anti-interference performance of the proposed approach. Five methods with the highest

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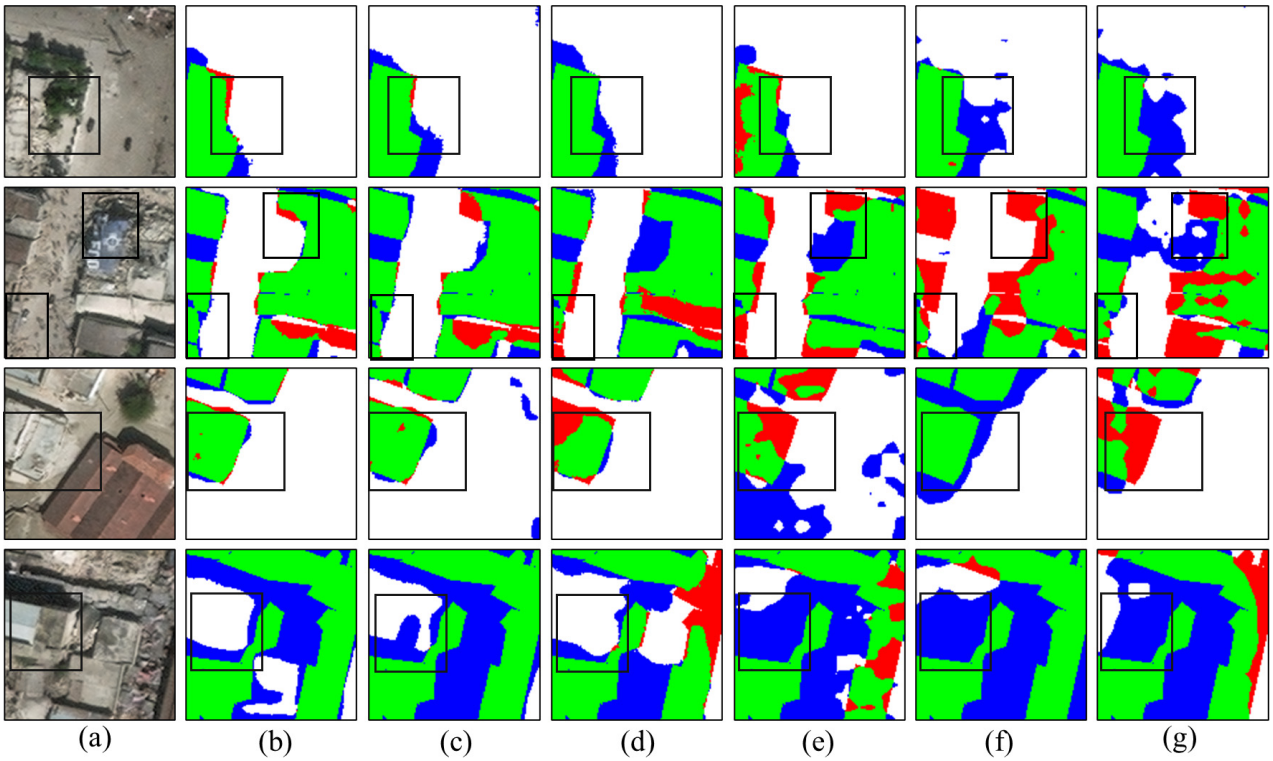

Fig. 13. Anti-interference analysis of our method versus other SOTA damaged building detection methods: (a) original image; (b) our method; (c) SIPE; (d) FDRNet; (e) OCNet; (f) DANet; and (g) ENCNet.

accuracy were selected for comparative analysis, including SIPE, FDRNet, object context network (OCNet), DANet, and context encoding network (ENCNet), as shown in Fig. 13. The first and second rows in Fig. 13 show images with common background objects that are similar to the damaged buildings in the remote sensing images, as shown in the black box. Although the spectral characteristics of these disturbed objects are similar to those of damaged buildings from a visual perspective, the damaged buildings have more complex texture and context information. Compared with the existing models, our method can avoid these interferences based on local–global context feature extraction.

The third and fourth rows include images of undamaged buildings and damaged buildings in the same region, as shown in the black box of Fig. 13. In the third line of Fig. 13, our method almost completely avoids the interference of undamaged buildings on the basis of ensuring the extraction of damaged buildings, which has obvious advantages over other methods. Especially in the fourth row, the scene environment is more complex, the false detection rate is higher, and other methods have a serious false detection rate. The OCNet, DANet, and ENCNet methods erroneously detect undamaged buildings and roads in the whole area as damaged buildings. The above analysis shows that our method has good resistance to various interferences in remote sensing images.

E. Transferability Analysis

To verify the transferability of our method, we established a challenging experiment. The Yushu earthquake in Qinghai, China, on April 14, 2011, was selected as the research object of our experiment. The magnitude of this earthquake was 7.1, with a focal depth of 14 km. There was a death toll of 2698, and 200 000 people were affected. The style of this area is quite different from that of the Haiti earthquake, which poses a great challenge to the transferability of our method. The data we selected are aerial images, and the images have a resolution of 0.5 m and consist of red, green, and blue bands. Some examples of the images and labels are shown in Fig. 14. In the same manner as the dataset in this article, we divided the image into damaged buildings and backgrounds. This dataset includes 642 images for training, 120 images for verification, and 165 images for testing, and the image size is $256 \times 256 \times 3$.

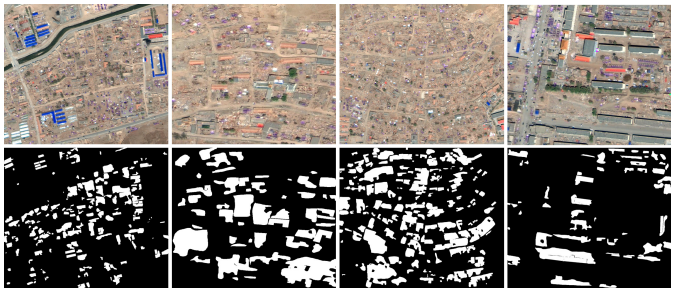

Fig. 14. Examples of images and labels from the Qinghai earthquake for transferability analysis.

1) Quantitative Analysis: We selected eight methods that performed well in the Haiti earthquake experiment for a transferability analysis on different datasets, including PSP-Net [33], Denseaspp [62], Deeplabv3_plus [63], ENCNet [64],

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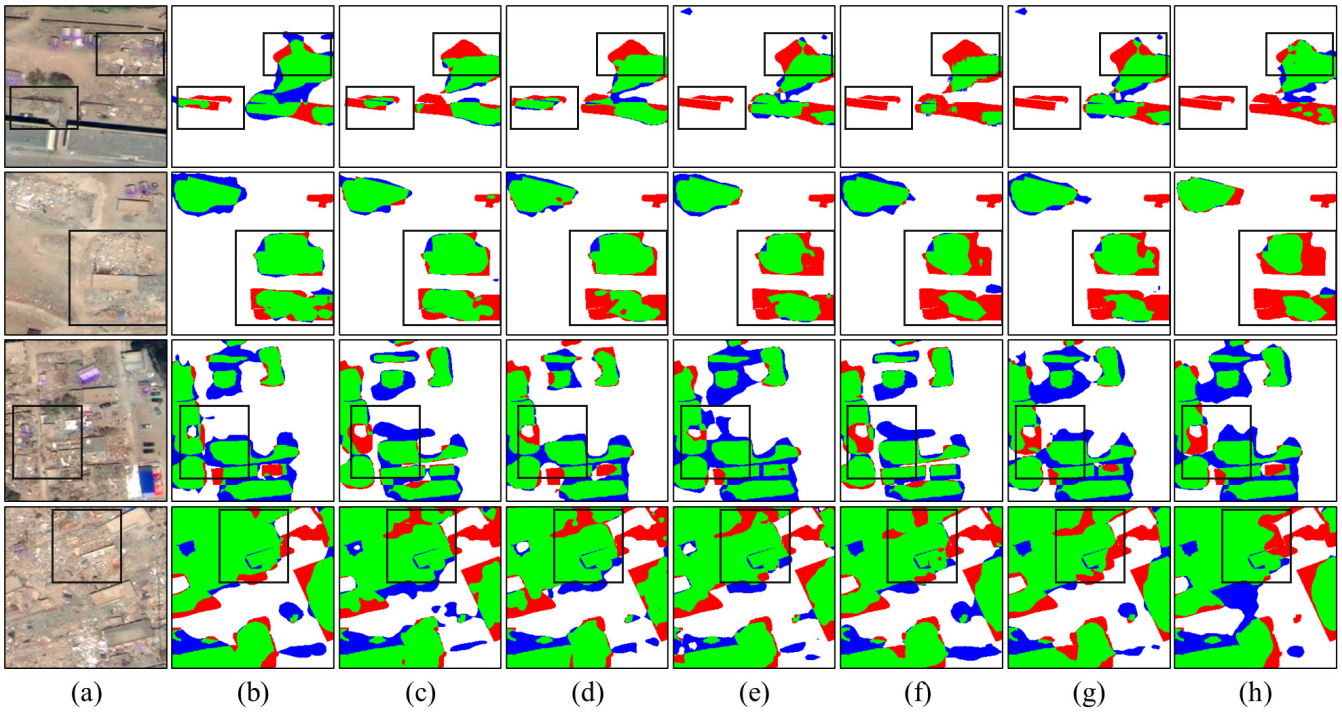

Fig. 15. Transferability analysis of our method versus other SOTA damaged building detection methods: (a) original image; (b) our method; (c) SIPE; (d) FDRNet; (e) ENCNet; (f) DANet; (g) OCNet; and (h) Deeplabv3_plus.

DANet [65], OCNet [66], FDRNet [67], and SIPE [68]. The results are shown in Table IV. It can be seen from Table IV that when using a small quantity of data, even after transfer learning, the features that the method can learn are still limited, and the accuracy of all methods was significantly reduced. Our method achieves high accuracy, with IOU, F1-score, and OA values reaching 62.43%, 76.11, and 85.87%, respectively. The main factor that determines the transferability of method portability is the ability of the network to extract features. In our method, the heterogeneity characteristics of damaged buildings are fully considered, and different modules are built to solve these problems. From the quantitative results, we can see that our method has obvious advantages of transferability over the existing SOTA models.

TABLE IV
DAMAGED BUILDING DETECTION RESULTS IN DIFFERENT EARTHQUAKE AREAS

Methods	IOU (%)	F1	OA (%)
Denseaspp [62]	58.72	74.03	83.87
PSPNet [33]	59.01	74.22	83.63
Deeplabv3_plus [63]	60.17	75.52	83.71
OCNet [66]	61.19	75.92	84.25
DANet [65]	61.24	75.97	84.27
ENCNet [64]	61.45	76.02	84.2
FDRNet[67]	61.79	**76.38**	85.27
SIPE [68]	62.15	76.27	**85.92**
Ours	**62.43**	76.11	85.87

2) Qualitative Analysis: In addition to quantitative analysis, to more intuitively reflect the advantages of our methods in extracting the results of damaged buildings, we selected a representative research area for visual display, the method include SIPE, FDRNet, ENCNet, DANet, OCNet, and Deeplabv3_plus, as shown in Fig. 15. The first–fourth rows in Fig. 15 show the study areas with different damage densities by different methods. The density of buildings in the first and second rows in Fig. 15 is low. For example, some buildings in the black box in the first row of Fig. 15 have not completely collapsed, and almost all methods have missed their detection. However, our method can detect some areas although they are incomplete. In the second row, the damaged buildings are basically collapsed, and our method treats the boundary of the collapsed buildings more finely (highlighted by the black boxes). The density of the building area in the third row is significantly higher than that in the first two rows. Because the image of the building debris leads to more false extraction, our method can obviously avoid the false positive extraction phenomenon (highlighted by the black boxes). In the fourth row, the building density is further increased, and there are many misclassified detection pixels. However, we can completely extract the damaged areas and avoid false negative extraction (highlighted by the black boxes). In general, most methods can extract the main features of the damaged buildings with different densities. However, compared with other methods, the details found using our method are more refined, and the extracted results are more complete.

F. Ablation Experiments

In this article, we establish a network suitable for damaged building detection according to the heterogeneity characteristics analysis of damaged buildings. The network has multiple

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TABLE V
DETAILED ABLATION EXPERIMENTS OF OUR METHOD

Methods	IOU (%)	P (%)	R (%)	F1	OA (%)
(1) Only Resnet101 in LGAM	65.59	75.01	83.93	79.22	84.92
(2) Without attention in LGAM	67.84	78.85	82.93	80.84	85.68
(3) Without FFS	66.74	77.53	82.75	80.05	85.20
(4) With one-scale prediction	64.80	73.8	84.15	78.64	84.64
(5) With four-scale prediction	67.76	77.95	82.91	80.35	85.25
Ours	68.87	79.78	82.31	81.03	85.83

complex modules and close connection structures. Therefore, to verify the contribution of each module to the proposed approach, we performed a detailed ablation experiment under the same data and experimental environment. According to the structure of the proposed approach, five groups of ablation experiments were carried out in this section.

1) *Only ResNet101 in LGAM*: To verify the role of the LGAM, we used only ResNet101 as the backbone in the encoding stage. As shown in Table V, compared with the complete LGAM structure, the IOU, F1-score, and OA are reduced by 3.28%, 1.81, and 0.91%, respectively. The structure of ResNet101 is widely used in the existing network model as a baseline structure with a good performance, but it lacks the correlation ability between detailed information and global information. In addition, the LGAM is established as a feature extraction structure with a stronger correlation between layers, which is also an obvious advantage of our method in the process of feature extraction.
2) *Without Attention in LGAM*: We further proved the role of attention in the encoding stage, which is shown in Fig. 5(b). In the ablation experiment, we only removed the attention module of Fig. 5(b) and established a feature extraction structure using the local detail awareness module and ResNet101. As shown in Table V, the IOU, F1-score, and OA are reduced by 1.03%, 0.19, and 0.15%, respectively. The decrease in accuracy is obvious, which further proves the importance of the proposed approach. Based on the local detail awareness module in Fig. 5(a), the attention module can further improve the global feature extraction ability.
3) *Without FFS*: In the stage of feature fusion between encoding and decoding, we used the commonly used feature map connection structure to replace the FFS structure (Fig. 6). Table V shows that the IOU, F1-score, and OA of the ablation experiment were reduced by 2.13%, 0.98, and 0.63%, respectively, which is extremely obvious. Although the connection between encoding and decoding has been proven to be important in previous research, the common connection structure lacks self-learning ability. In this article, the structure of FFS has self-learning ability while connecting the characteristics of the encoding and decoding stages, which is necessary to improve the network performance.
4) *With One-Scale Prediction*: To prove the effectiveness of the multibranch auxiliary classifier, we first carried out a comparative experiment through a one-scale output. In this experiment, only the output of the last layer was retained, and the result is shown in Table V. Compared with the proposed approach, the IOU, F1-score, and OA were reduced by 4.07%, 2.39, and 1.19%, respectively, which is also the group with the worst performance in all ablation experiments. This experiment further proves the importance of multiscale output for the network.
5) *With Four-Scale Prediction*: Based on one-scale prediction, we fused four scales to obtain the final prediction results, including Predict1, Predict2, Predict3, and Predict4 (Fig. 7). This structure was also the most used in the multiscale prediction of semantic segmentation tasks. Table V shows that compared with the four-scale output, the proposed classifier increases the IOU, F1-score, and OA by 1.11%, 0.68, and 0.58%, respectively. In addition, the proposed multibranch auxiliary classifier only added a branch prediction structure based on the original network, which does not increase additional computation and time. This experiment also proves that the multibranch auxiliary classifier is necessary in the process of final prediction.

G. Advantages and Limitations of the Proposed Approach

1) Advantages of the Proposed Approach: The proposed approach has three attractive advantages. First, this method is established based on analyzing the heterogeneity characteristics of damaged buildings in remote sensing images. In addition, it is proven that our method has high accuracy and good stability. Second, compared with traditional methods, in our method, features do not need to be manually designed and specific classifiers do not need to be selected, which can effectively improve the degree of automation and detection efficiency. Finally, our method only needs to input the remote sensing image after an earthquake and does not need other auxiliary data, which is more direct and simpler in terms of data acquisition. In addition, the whole model can be added to any existing emergency rescue system as a separate embedded module, which can be better applied to emergency rescue after an earthquake.

2) Limitations of the Proposed Approach: Although the experimental results show that our method has obvious advantages over the existing model for damaged building detection, there are still some shortcomings. First, the deep learning model requires a large amount of data, and the amount of data in this experiment is limited. Although the effectiveness of our method is proven from many aspects, the amount of data still needs to be further increased. Second,

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although our method is established based on the heterogeneous characteristics of damaged buildings, the training data are limited to images from Haiti. In the real world, earthquakes of different geographical locations and damage levels will lead to different characteristics, which will also lead to a decline in the performance of the method when directly used with other areas. The universality of our method still needs further research. Finally, the generation of earthquakes is often accompanied by inclement weather, a single source of remote sensing images may have feature losses, and the detection of damaged buildings by multisource image fusion should also be further considered.

VI. Conclusion

We have proposed a model for damaged building detection from remotely sensed images considering heterogeneity characteristics in this article. First, an LGAM is constructed to extract the characteristics of damaged buildings as much as possible by analyzing the heterogeneity characteristics of damaged buildings. Second, the structure of the FFS is constructed, which can fully integrate the characteristics of different stages and improve the restoration ability of detailed features in the process of continuous upsampling. A multibranch auxiliary classifier structure is established to aggregate semantic and detailed features at different scales in the decoding stage. The experimental results show that the proposed approach has obvious advantages over the SOTA models in recent years. In addition, the advantages of our method are verified by the analysis of many aspects, and the final ablation experiments also prove the role of each module in the proposed model. Finally, the advantages and disadvantages of our method are analyzed in detail.

We will conduct in-depth research on this basis in the future. First, we will consider using multisource remote sensing images to extract features from damaged buildings for better accuracy after an earthquake. Second, the applicability of our method in different geographical environments is also a considerable challenge. Finally, we will further study different types of damage and statistical data after an earthquake to realize a complete disaster evaluation system and provide a data basis for subsequent rescue, insurance, claim settlement, and other applications.

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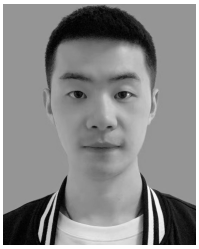

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